

Updated ER Diagram

University Enrolment & Lecture Management System

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MSE800 — Week 3, Activity 4

Relationships

The model contains two relationships. Both are *ternary* relationships in the source diagram; each is described below together with its cardinalities.

Enrolls — Student : Enrollment : Lecture. A **Student** *enrolls in* a **Lecture** through an **Enrollment** record. This is a *many-to-many* association between *Student* and *Lecture*, resolved by the associative entity *Enrollment*:

- one *Student* may have *many* enrollments (M);
- one *Lecture* may be taken by *many* students (M);
- each *Enrollment* links exactly *one* student to exactly *one* lecture (1).

Lectures — Lecturer : Lecture : Subjects. A **Lecturer** *delivers* a **Lecture** that belongs to a particular **Subject**. This ternary relationship decomposes into two *one-to-many* associations that meet at *Lecture*:

- one *Lecturer* delivers *many* lectures ($1 : N$);
- one *Subject* has *many* lectures ($1 : N$);
- each *Lecture* is delivered by exactly *one* lecturer and belongs to exactly *one* subject (1).

Changes from the original diagram

- **Added** an **Email** attribute to *Student*, **Credits** and **Description** attributes to *Subjects*, a **Room** attribute to *Lecture*, and a **Grade** attribute to *Enrollment*.
- **Corrected** the *Lecturer* key: **Lecture_id** becomes **Lecturer_id** (a lecturer is identified by their own id, not by a lecture).
- **Corrected** the *Enrollment* key: **Student_code** alone cannot identify an enrollment (a student may enrol in several courses), so a surrogate **Enrollment_id** primary key is introduced.
- **Resolved** the overlapping course attributes (**CC#** / **Subject** / **Subject_code**) into a single foreign key to the *Subjects* entity.

ER Diagram

Legend: rectangles are entities, diamonds are relationships, ellipses are attributes, and underlined attributes are primary keys.

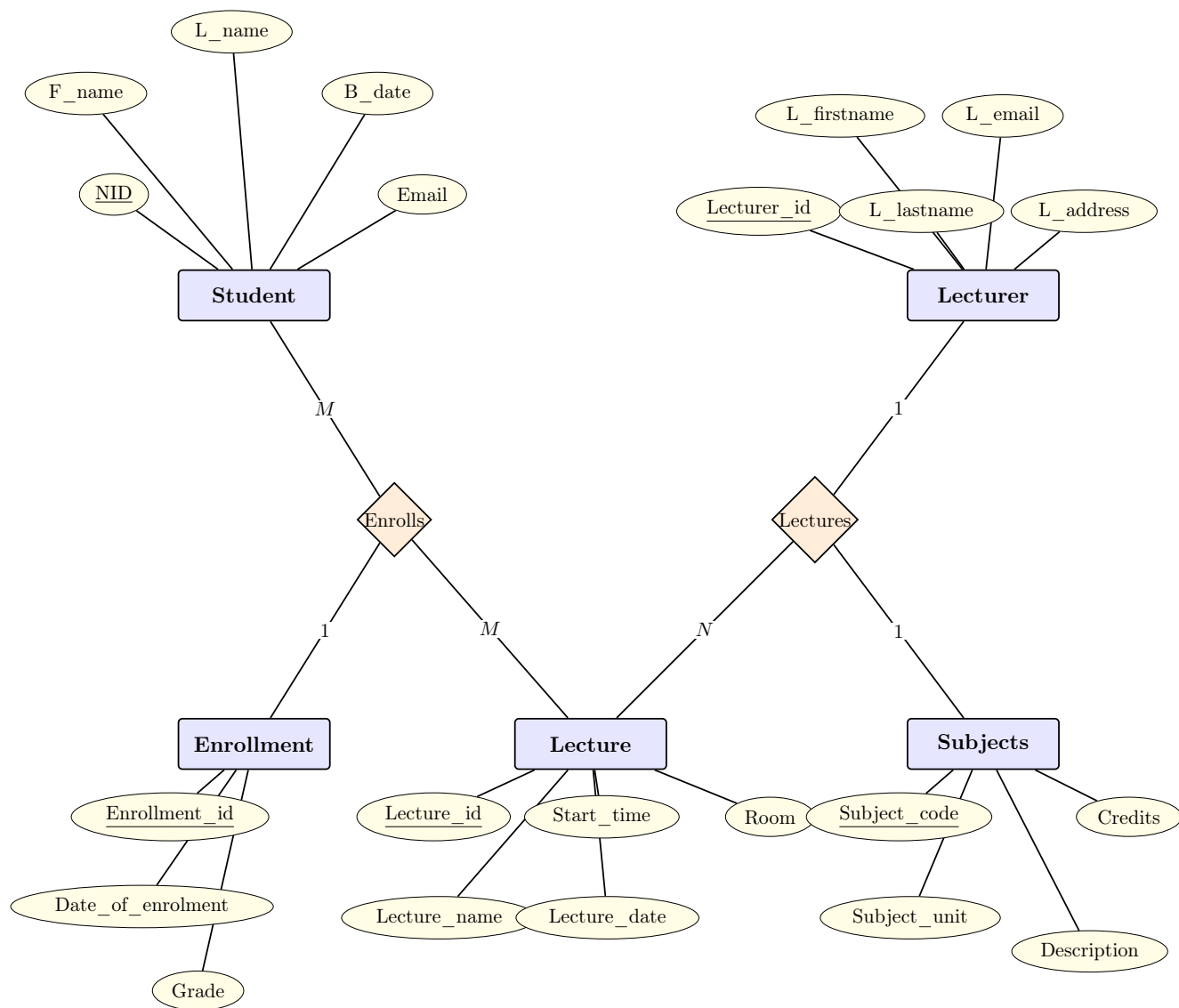


Figure 1: Updated ER diagram in Chen notation. Primary keys are underlined. Foreign keys are carried by the relationships rather than shown as attributes.